

DANE TIPENE

Analytics & Automation Engineer

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PROFESSIONAL SUMMARY

I walk into broken, expensive, manual processes and automate them out of existence. At the Essential Services Commission I've delivered enterprise-grade automation inside one of Victoria's most regulated government environments — building the organisation's first locally-deployed, multi-model AI pipeline and repeatedly eliminating weeks of manual effort through R automation and intelligent workflow design. I build systems that work without handholding, scale without friction, and make data easy to act on for the people who matter.

CORE COMPETENCIES

- **Data Automation & Process Optimisation:** R automation scripting, Python workflows, workflow scheduling, process monitoring
- **Business Intelligence & Visualisation:** Tableau, executive dashboards, data storytelling, stakeholder reporting
- **Database & Workflow Technologies:** SQL, Azure SQL environment, workflow scheduling, data quality processes
- **Advanced Analytics:** Python (Pandas, NumPy, Scikit-Learn), SQL, statistical analysis, exploratory data analysis
- **Machine Learning:** XGBoost, Random Forest, regression models, feature engineering, predictive modelling, model evaluation
- **Data Quality & Governance:** Regulatory compliance, data integrity frameworks, audit trails, automated QA processes
- **Independent Execution:** Autonomous project delivery, self-directed problem-solving, comprehensive documentation

PROFESSIONAL EXPERIENCE

Data Analyst, Essential Services Commission | May 2025 – Present | Melbourne, VIC

- Built automated QA pipelines in R that reduced the quarterly regulatory data validation process from 12+ days of manual work to under 3 days, strengthening data integrity, auditability, and team capacity
- Engineered enterprise-grade automated reporting system in R (5,000+ lines, orchestrator pattern with 10+ modular functions) that reduced Retailer Profile Report generation from 30+ days to 10 minutes, integrating Azure SQL database queries, dynamic calculations, and automated visualization generation to produce 30+ interactive HTML reports with intelligent, scenario-based commentary
- Designed and deployed an in-house transcription pipeline processing ~3,500 enforcement call recordings — locally hosted using WhisperX and three Pyannote neural models for speaker diarisation, orchestrated through R and connected to SharePoint. Sensitive audio never left the team's infrastructure. Established the Commission's first locally deployed, multi-model AI pipeline within existing IT and AI policy boundaries.
- Own monthly disconnections reporting providing critical regulatory insights to Energy Division leadership, CEO, and Commissioners, directly informing consumer protection policy decisions
- Execute and maintain quarterly license fee calculation scripts that determine Commission funding structure, implementing comprehensive QA processes to ensure financial accuracy and regulatory compliance

Data Analyst (Contract), Waikare Community Development & Research Trust | July – October 2024 | HRCNZ Funded Project

Pona Kaká Initiative

- Conducted comprehensive qualitative and quantitative analysis on Māori arthritis management for the Pona Kaká Initiative, working independently to transform interview transcripts into structured, actionable insights
- Built analysis-ready datasets independently using Python and Excel, performing descriptive statistics and trend analysis that informed community health recommendations and policy discussions

- Created stakeholder-ready Canva reports that translated complex research findings into accessible formats, earning recognition from project sponsors and invitation for future collaborations

Retail Forex Trader, Self Employed | May 2020 – January 2024 | *Location Independent*

- Managed personal trading portfolio, analysing large financial datasets to optimise strategies—improving win/loss performance by 31% through disciplined, data-driven decision-making
- Implemented systematic backtesting and risk management frameworks, reducing trading losses by 25% through rigorous strategy validation and continuous performance monitoring
- Led strategy workshops for traders, demonstrating ability to communicate complex analytical insights effectively and mentor others in data-driven approaches

MACHINE LEARNING PORTFOLIO

Employee Attrition Prediction Model | **Google Advanced Data Analytics Professional Certificate** | February 2025

- Built XGBoost predictive model achieving 96.7% AUC-ROC for employee attrition forecasting, enabling proactive HR retention strategies through data-driven insights
- Improved recall by 15.38% through strategic feature engineering and data leakage elimination, demonstrating rigorous analytical methodology and model optimisation skills
- Developed comprehensive, actionable recommendations for HR teams to implement retention strategies based on workload balance, career growth, and compensation transparency

TikTok Content Moderation Classification System | **Google Advanced Data Analytics Professional Certificate** | January 2025

- Developed Random Forest classification model achieving 99.5% accuracy and 99.21% recall for automated content moderation—dramatically reducing manual review workload while maintaining quality standards
- Conducted comprehensive exploratory data analysis using Python (Pandas, NumPy, Seaborn) to uncover engagement patterns and inform model development
- Created interactive Tableau dashboards enabling stakeholders to explore key findings and data-driven insights independently

LEADERSHIP & OPERATIONAL EXCELLENCE

22 years leading high-performance teams in hospitality management, including 3 Head Chef positions. Consistently delivered results under pressure while managing operations, budgets, and team development. Reduced operational waste by 30% through data-driven decision-making. This foundation in autonomous leadership, accountability, and systematic problem-solving directly informs analytical approach and independent execution capability.

EDUCATION

- **Professional Certificate** | **Google Business Intelligence**, Coursera — *In Progress Developing expertise in ETL pipeline design, data warehouse architecture, and advanced BI dashboard development*
- **Professional Certificate** | **Google Advanced Data Analytics**, Coursera — *Completed 2025*
- **Professional Certificate** | **Google Data Analytics**, Coursera — *Completed 2024*
- **Course** | **Google AI Essentials**, Coursera — *Completed 2024*
- **Professional Cookery Levels 1 – 4** | *Hospitality Standards Institute*